

Numerical Method: Newton-Raphson Multivariate

$$\text{Let } (x_0, y_0) = (0, 0.5)$$

$$(x_1, y_1) = (2, 3.5)$$

$$(x_2, y_2) = (1, 1.5)$$

$$(x_3, y_3) = (3, 7.5)$$

Fitting to a line:

$$\begin{aligned} e(x) &= y^*(x) - y_i(x_i) \\ &= c_0 + c_1 x - y_i(x_i) \end{aligned}$$

$$P = (c_0 - 0.5)^2 + (c_0 + 2c_1 - 3.5)^2 + (c_0 + c_1 - 1.5)^2 + (c_0 + 3c_1 - 7.5)^2$$

$$\text{MSE} = \frac{1}{4} \sum_{i=1}^4 ((c_0 + c_1 x_i) - y_i)^2$$

sub in each pair into the MSE formula.

$$\frac{\partial \text{MSE}}{\partial c_1} = 2(c_0 + 2c_1 - 3.5)(2) + 2(c_0 + c_1 - 1.5)(1) + 2(c_0 + 3c_1 - 7.5)(3)$$

$$\frac{\partial \text{MSE}}{\partial c_1^2} = 4(2) + 2(1) + 6(3) = 28$$

$$\frac{\partial \text{MSE}}{\partial c_0} = 2(c_0 - 0.5)(1) + 2(c_0 + 2c_1 - 3.5)(1) + 2(c_0 + c_1 - 1.5)(1) + 2(c_0 + 3c_1 - 7.5)(1)$$

$$\frac{\partial \text{MSE}}{\partial c_0^2} = 2 + 2(1) + 2(1) + 2(1) = 8$$

Hold c_0 as constant and optimize for c_1 with Newton-Raphson.

Set $c_0 = 0$ and $c_1 = 0$ (initial guess):

$$c_1^* = 0 - \frac{4(0 + 2(0) - 3.5) + 2(0 + 0 - 1.5) + 6(0 + 3(0) - 7.5)}{28} \quad \text{Iteration 1}$$

$$= 3\frac{1}{4} \approx 2.214$$

$$c_1^* = 2.214$$

Now set $c_1 = 2.214$ as constant and optimize for c_0 .

Now hold C_1 as constant and optimize for C_0 with Newton-Raphson.

Set $C_1 = 31/4$ and $C_0 = 0$ (initial guess):

$$C_0^* = \frac{31/4 - 2(0 - 0.5) + 2(0 + 2(31/4) - 3.5) + 2(0 + 31/4 - 1.5) + 2(0 + 3(31/4) - 7.5)}{8}$$

$$= -0.0714$$

Iteration 1

Now hold C_0 as constant (-0.0714) and optimize for C_1 with Newton-Raphson.

$$C_1^* = 2.214 - \frac{4(-0.0714 + 2(2.214) - 3.5) + 2(-0.0714 + 2.214 - 1.5) + 6(-0.0714 + 3(2.214) - 7.5)}{28}$$

$$= 2.245$$

$$C_1 = 2.245$$

Iteration 2

Now hold C_1 as constant and optimize for C_0 with Newton-Raphson.

Set $C_1 = 2.245$ and $C_0 = -0.0714$.

$$C_0^* = -0.0714 - \frac{2(-0.0714 - 0.5) + 2(-0.0714 + 2(2.245) - 3.5) + 2(-0.0714 + 2.245 - 1.5) + 2(-0.0714 + 3(2.245) - 7.5)}{8}$$

$$C_0 = -0.1174$$

Best line of fit: $y = -0.1174 + 2.245x$

Fitting to a curve:

$$\text{Let } (x_0, y_0) = (0, 0.5)$$

$$(x_1, y_1) = (2, 3.5)$$

$$(x_2, y_2) = (1, 1.5)$$

$$(x_3, y_3) = (3, 7.5)$$

$$e(x) = y(x) - y_i^*(x_i)$$

$$= Bx^2 + Cx + D - y_i(x_i)$$

$$\rho = (D - 0.5)^2 + (4B + 2C + D - 3.5)^2 + (B + C + D - 1.5)^2 + (9B + 3C + D - 7.5)^2$$

$$\text{MSE} = \frac{1}{4} \sum_{i=1}^4 \sqrt{(Bx^2 + Cx + D) - y_i^2}$$

$$\frac{\partial \text{MSE}}{\partial B} = 2(4B+2C+D-3.5)(4) + 2(B+C+D-1.5)(1) + 2(9B+3C+D-7.5)(9)$$

$$\frac{\partial \text{MSE}}{\partial B^2} = 8(4) + 2(1) + 18(9) = 196$$

$$\frac{\partial \text{MSE}}{\partial C} = 2(4B+2C+D-3.5)(2) + 2(B+C+D-1.5)(1) + 2(9B+3C+D-7.5)(3)$$

$$\frac{\partial \text{MSE}}{\partial C^2} = 4(2) + 2(1) + 6(3) = 28$$

$$\frac{\partial \text{MSE}}{\partial D} = 2(D-0.5) + 2(4B+2C+D-3.5) + 2(B+C+D-1.5) + 2(9B+3C+D-7.5)$$

$$\frac{\partial \text{MSE}}{\partial D^2} = 2 + 2 + 2 + 2 = 8$$

Hold C and D constant (=0), optimize B.

Let initial guess $B = 0$.

$$B^* = 0 - \frac{8(-3.5) + 2(-1.5) + 18(-7.5)}{196} = 0.847$$

$$B^* = 0.847.$$

Hold B and D constant (=0.847, =0), optimize C.

Let initial guess $C = 0$.

$$C^* = 0 - \frac{4(4(0.847)-3.5) + 2(0.847-1.5) + 6(9(0.847)-7.5)}{28} = 0.036$$

$$C^* = 0.036$$

Hold B and C constant (=0.847, 0.036), optimize D.

$$D^* = 0 - \frac{2(-0.5) + 2(4(0.847) + 2(0.036) - 3.5) + 2(0.847 + 0.036 - 1.5) + 2(9(0.847) + 3(0.036) - 7.5)}{8}$$

$$D^* = 0.2315$$

Iteration 1

Iteration 1

Iteration 2

Hold C and D constant (= 0.036, 0.2315) and optimize B.

$$B = 0.847$$

$$B^* = 0.847 - \frac{8(4(0.847) + 2(0.036) + 0.2315 - 3.5) + 2(0.847 + 0.036 + 0.2315 - 1.5) + 18(9(0.847) + 3(0.036) + 0.2315 - 7.5)}{196}$$

Iteration 2

$$B = 0.8$$

Hold B and D constant (= 0.8, 0.2315), optimize C.

$$C = 0.036$$

$$C^* = 0.036 - \frac{4(4(0.8) + 2(0.036) + 0.2315 - 3.5) + 2(0.8 + 0.036 + 0.2315 - 1.5) + 6(9(0.8) + 3(0.036) + 0.2315 - 7.5)}{28}$$

Iteration 2

$$C = 0.058$$

Hold B and C constant (= 0.8, 0.058), optimize D.

Iteration 2

$$D^* = 0.2315 - \frac{2(0.2315 - 0.5) + 2(4(0.8) + 2(0.058) + 0.2315 - 3.5) + 2(0.8 + 0.058 + 0.2315 - 1.5) + 2(9(0.8) + 3(0.058) + 0.2315 - 7.5)}{8}$$

$$D = 0.363$$

Best curve of fit: $y = 0.8x^2 + 0.058x + 0.363$